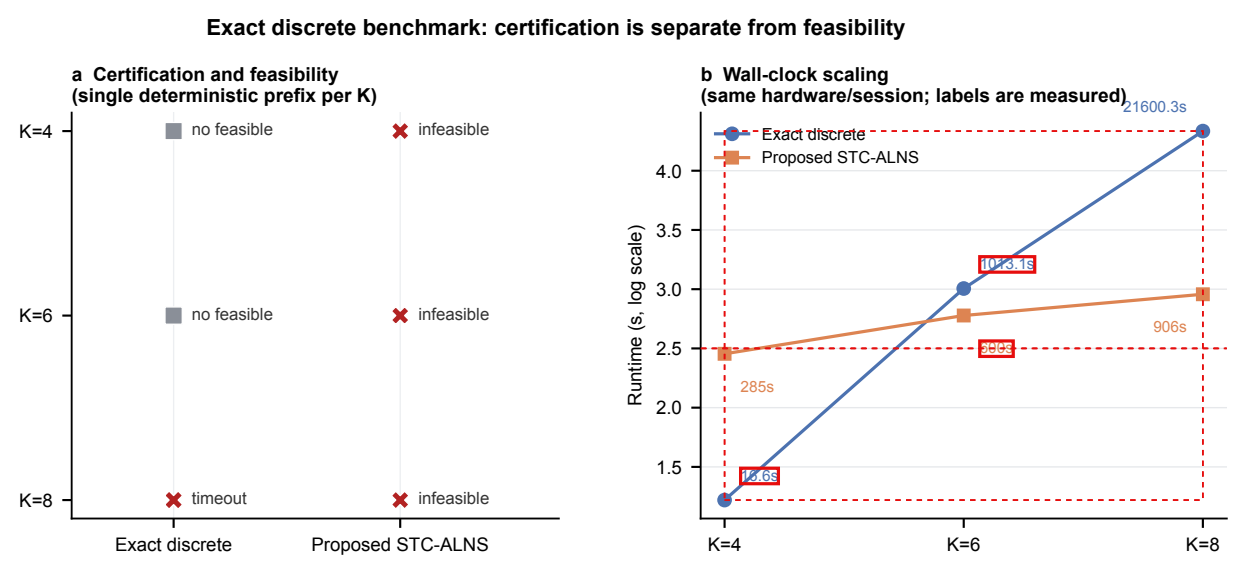
Exact discrete benchmark: certification is separate from feasibility
a Certification and feasibility
(single deterministic prefix per K)
K=4
K=6
K=8
no feasible
no feasible
infeasible
infeasible
timeout
infeasible
Exact discrete
Proposed STC-ALNS
b Wall-clock scaling
(same hardware/session; labels are measured)
Exact discrete
Proposed STC-ALNS
Runtime (s, log scale)
4.0
3.5
3.0
2.5
2.0
1.5
21600.3s
1013.1s
906s
285s
12.6s
K=4
K=6
K=8